

Response to Reviewers

Manuscript ID: BIOPHYSICAL-JOURNAL-D-25-00813R1

Manuscript title: "AFM-Fold: Rapid Reconstruction of Protein Conformations from AFM Images"

We sincerely thank the reviewers for their positive evaluation of the revised manuscript and for the remaining minor comments. Because the current submission is for final publication materials, we provide a concise summary of the final clarifications made in the manuscript.

Reviewer 1

We thank the reviewer for confirming that the previous concerns have been addressed satisfactorily and that the manuscript is suitable for publication. No additional changes were requested.

Reviewer 2

We thank the reviewer for the positive assessment and for pointing out the remaining issue with Figure 5. We corrected the Figure 5 legend so that the panel descriptions match the figure layout. We also added a brief clarification that the fitted orientations shown in the representative cases are independently optimized for image comparison and should not be interpreted as physical reorientations of the molecule on the substrate during HS-AFM imaging.

Reviewer 3

We thank the reviewer for the constructive minor clarifications. In the final manuscript and Supplemental Material, we clarified that the FlhAC schedule sweep used a reduced two-CV subset, whereas the final inference used the three-CV set reported in the main text and Supplemental Table S3. We also moderated the broad "for the first time" statement and harmonized the CNN terminology by referring to the architecture as a group-equivariant CNN while noting that its group-pooled output is invariant where relevant.

We are grateful to the reviewers and the Editorial Office for their careful evaluation of our work.